

TABLE 4
THE STATUS OF MAJOR SEMICONDUCTOR MANUFACTURING PROJECTS IN INDIA

What	Under	Where	Who	Start	How Much (INR)
Fab	India Semiconductor Mission	Dholera, Gujarat	Tata Electronics, PSMC	29 February 2024	910B
OSAT unit	India Semiconductor Mission	Morigaon, Assam	Tata Semiconductor Assembly and Test	29 February 2024	270B
OSAT unit	India Semiconductor Mission	Sanand, Gujarat	CG Power, Renesas Electronics (Japan) Stars Microelectronics (Thailand)	29 February 2024	76B
ATMP unit	India Semiconductor Mission	Sanand, Gujarat	Micron Technology	June 2023	225.16B
ATMP unit	India Semiconductor Mission	Sanand, Gujarat	Kaynes Semicon	2 September 2024	33B
Fab	MH gov, ISM (pending)	Taloja, Panvel, Maharashtra	Tower Semiconductor (Israel), Adani Group	5 September 2024	839.47B
OSAT facility	MH gov, Sachin Tendulkar	Navi Mumbai, Maharashtra	RRP Electronics, HMT Microelectronic AG	September 2024	365.73B
OSAT facility	GJ gov, ISM (pending)	Surat, Gujarat	Suchi Semicon	Opened in Dec 2024	\$100M

Even if European and American fabs increase production, the supply chain remains lengthy and expensive.

And this is precisely where India comes in.

The advent of India and why India

According to *Reuters*, India's semiconductor market is projected to reach \$63 billion by 2026, signalling its growing influence in the global chip economy.

Global supply chain disruptions and escalating geopolitical tensions have prompted semiconductor companies to reconsider their manufacturing and investment strategies. Despite the European Union's €43 billion Chips Act, both the US and EU still lack substantial manufacturing capacity, making them dependent on external sources.

Meanwhile, East Asia continues to dominate semiconductor production, with Taiwan, South Korea, Japan, and China accounting for over 70% of global chip manufacturing.

But this concentration comes with risks. Taiwan's geopolitical vulnerability, tensions between the US and China, and potential disruptions in South Korea and Japan have fueled concerns about supply chain security. Countries are now actively

looking for alternative and 'friendly' manufacturing destinations, and India might be an attractive option.

The country offers cost-effective labour, a large talent pool of English-speaking engineers, and favourable government policies. As Sarah E. Kreps, Professor at Cornell University, in a *Wired* report, points out:

"India has always had the potential, but until now, other semiconductor hubs have met the demand. This is India's moment to capitalise on its strengths."

Government initiatives and the talent pool

The Indian government is well aware of this "strength." It has responded to these opportunities with aggressive policy measures, like the India Semiconductor Mission (ISM). Union Minister Ashwini Vaishnaw even announced that the first 'Made-in-India' chips would be produced by mid-2025.

India's semiconductor strategy is manufacturing chips and creating a holistic ecosystem that includes design, testing, packaging, and research. The country already has a strong foothold in semiconductor design, with companies like Intel, AMD, and Qualcomm operating large R&D centres in Bengaluru and Hyderabad. The next step is to

convert design expertise into large-scale manufacturing.

The India Brand Equity Foundation (IBEF) highlights several reasons why India is emerging as a key semiconductor hub. India's strengths include a cost-effective workforce and a vast pool of skilled engineers, making it an attractive semiconductor hub. Strategic policies, such as the ₹760 billion Production Linked Incentive (PLI) scheme, encourage chip production, while the Design Linked Incentive (DLI) scheme supports domestic design startups. Additional measures include special economic zones (SEZs), semiconductor parks, and duty exemptions.

Competing with East Asia: Can India catch up?

Despite its advantages, India faces stiff competition from well-established semiconductor giants like Taiwan and South Korea. TSMC, Samsung, and Intel continue to expand their production in Asia and the US, and China is investing over \$150 billion in semiconductor development.

However, the global semiconductor landscape is not just about advanced chip manufacturing. Many industries—such as automotive, industrial, and telecom—still rely on mature node chips, an area